

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Database Management Systems
COURSE CODE: CSA0501

COURSE OUTCOME COVERED

CO1: *Analyze DBMS architecture, different data models, schema types, and design ER/EER diagrams, relational schemas for case-based scenarios by identifying entities and relationships. (BL1, BL2)*

Activity Tool : Technical Presentation (Low)
Assessment Tool Weightage: 35%

Dr.A.JAYA MABEL RANI

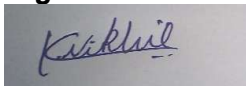
QUESTIONS		Total Marks: 50	
Q.No	Question	Bloom's Level	Marks
1	Prepare a 5-slide presentation introducing DBMS: definition, objectives, advantages, and real-world applications.	BL2 – Understand	5
2	Present a comparison between file-based systems and DBMS using real-life case studies from banking or healthcare.	BL2 – Understand	5
3	Deliver a presentation explaining the three-level ANSI/SPARC architecture using diagrams and practical examples.	BL2 – Understand	5
4	Present the different data models (Hierarchical, Network, Relational) with diagrams and industry use cases.	BL1 – Remember	5
5	Create and present an ER diagram for an Online Library System, explaining each component clearly.	BL2 – Understand	5
6	Prepare a presentation on the EER model, demonstrating generalization and specialization with a University case study.	BL2 – Understand	5
7	Present the concept of data independence (logical and physical) and its significance in DBMS architecture.	BL2 – Understand	5

8	Deliver a technical presentation on the software components of a DBMS, explaining query processor, storage manager, and transaction manager.	BL1 – Remember	5
9	Prepare a presentation comparing strong and weak entities in ER modeling with real-world examples.	BL2 – Understand	5
10	Present a complete design walkthrough of an EER diagram for a Retail Chain Management System.	BL2 – Understand	5

Code of Conduct:

I K.NIKHIL / 192424366 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



RUBRICS FOR CALCULATION:

Technical Presentation					
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals

Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Topics Covered

1 Introduction to DBMS

2 File-Based System vs DBMS

3 ANSI/SPARC Architecture

4 Different Data Models

5 Online Library ER Diagram

6 University EER Model

7 Data Independence

8 DBMS Software Components

9 Strong and Weak Entities

10 Retail Chain EER Design

Introduction to DBMS



What is a DBMS?

- DBMS stands for Database Management System
- Software used to create and manage databases
- Stores, organizes, retrieves, and updates data
- Acts as an interface between users and databases
- Helps manage large amounts of data efficiently



Popular Examples

- MySQL
- Oracle
- PostgreSQL
- Microsoft SQL Server

Objectives and Advantages



Objectives

- Reduce data redundancy
- Maintain data consistency
- Provide data security
- Support data sharing
- Ensure data integrity
- Provide backup and recovery



Advantages

- Easy data access
- Faster data retrieval
- Multiple-user support
- Controlled data access
- Better data management

Applications of DBMS



Real-World Use Cases

- Banking — customer accounts and transactions
- Healthcare — patient records and prescriptions
- Education — student details, marks, and attendance
- E-Commerce — products, orders, and payments
- Railway / Airline — ticket booking and seat management

QUESTION 2

File-Based System vs DBMS

Traditional File-Based System



Characteristics

- Data is stored in separate files
- Different departments maintain separate records
- Data may be duplicated in multiple files
- Updating data is difficult
- Data sharing is limited



Example Files

- Customer.txt
- Account.txt
- Loan.txt
- Transaction.txt
- Problem: the same customer info may repeat across many files

Comparison: File System vs DBMS

Feature	File-Based System	DBMS
Data storage	Separate files	Centralized database
Redundancy	High	Low
Security	Limited	Strong
Data sharing	Difficult	Easy
Backup	Manual	Supported
Multi-user access	Limited	Supported
Data retrieval	Slow	Fast

Real-Life Case Study



Banking

- File system: customer info repeated in account and loan files
- DBMS: customer data stored once, linked to all services



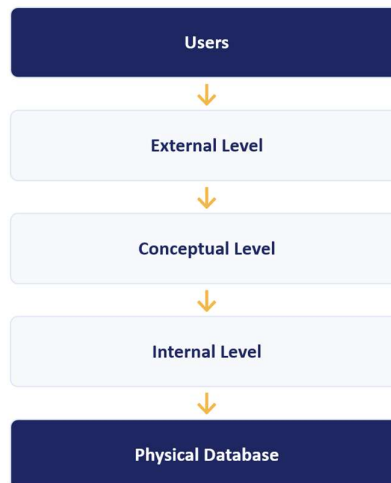
Healthcare

- File system: separate patient records per department
- DBMS: records connected across registration, lab, pharmacy, billing

QUESTION 3

ANSI/SPARC Architecture

Three-Level DBMS Architecture



External and Conceptual Levels

E

External Level

- Represents the user view
- Different users see different information
- Improves security
- Student sees marks & attendance
- Teacher sees student & course details
- Administrator has complete access

C

Conceptual Level

- Represents the complete logical database
- Defines entities, attributes, and relationships

Internal Level and Benefits

I

Internal Level

- Describes how data is physically stored
- Manages files, indexes, storage blocks, and memory

B

Benefits of the Architecture

- Provides data abstraction
- Supports data independence
- Improves data security
- Allows multiple user views
- Simplifies database maintenance

QUESTION 4

Data Models

Types of Data Models

D

A Data Model Defines How Data Is:

- Organized
- Stored
- Connected
- Accessed

3

Main Types

- Hierarchical Model
- Network Model
- Relational Model

Hierarchical and Network Models

H

Hierarchical Model

- Uses a tree structure
- Follows parent-child relationships
- Supports one-to-many relationships
- Example: organization structure

N

Network Model

- Uses a graph structure
- Supports many-to-many relationships
- A record can have multiple parents
- Example: transportation and telecom

Relational Data Model

Model	Structure
Hierarchical	Tree
Network	Graph
Relational	Tables

Relational model: stores data in tables (rows & columns); uses primary and foreign keys; supports SQL; easy to understand and manage.

QUESTION 5

Online Library ER Diagram

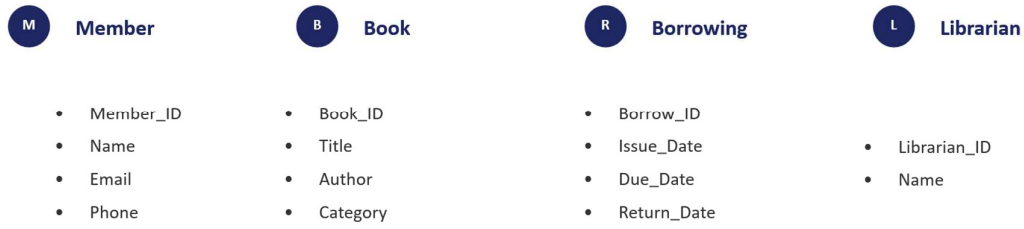
ER Diagram — Basic Components

E

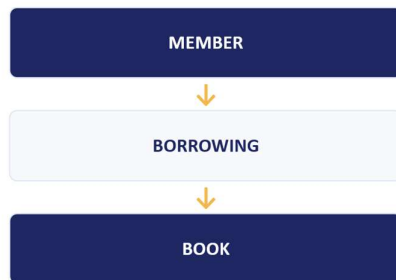
Core Building Blocks

- Entity — a real-world object
- Attribute — a property of an entity
- Relationship — a connection between entities
- Primary Key — a unique identifier
- Example: STUDENT (Student_ID, Name, Department)

Online Library System — Entities



Online Library ER Design



- One member can borrow many books
- One book can appear in many borrowing records
- A librarian manages borrowing activities
- Cardinality: MEMBER 1 : M BORROWING
- Cardinality: BOOK 1 : M BORROWING

QUESTION 6

University EER Model

Enhanced Entity-Relationship Model

+

EER Extends the ER Model With:

- Generalization
- Specialization
- Superclass
- Subclass
- Inheritance

P

Purpose

- Represents complex real-world systems more clearly

Generalization and Specialization



Generalization

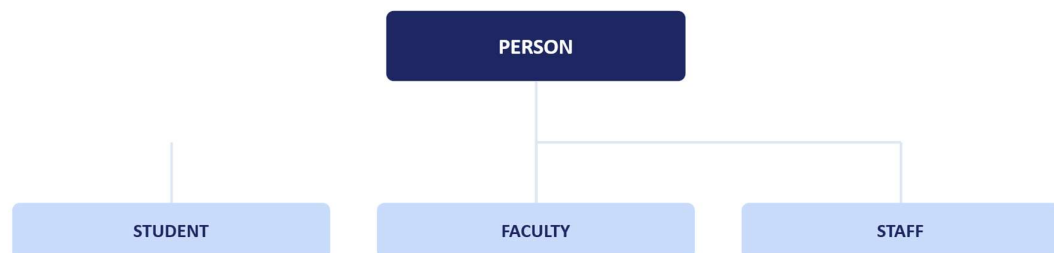
- Combines similar entities into one common entity
- Student + Faculty + Staff → Person



Specialization

- Divides one general entity into specialized entities
- Person → Student, Faculty, Staff

University EER Model



- Department
- Semester

- Subject
- Salary

- Job Role
- Salary

Common attributes (all subclasses): Person_ID, Name, Email, Phone

QUESTION 7

Data Independence

Data Independence



Definition

- The ability to modify one database level without affecting other levels

Types

- Logical Data Independence
- Physical Data Independence

Importance

- Reduces application changes
- Improves flexibility
- Simplifies maintenance

Logical Data Independence

L

How It Works

- Changes occur at the conceptual level
- User views and application programs remain unaffected

E

Example

- Before: Student_ID, Name, Department
- After: Student_ID, Name, Department, Email
- Adding Email should not affect existing applications

Physical Data Independence

Logical	Physical
Changes logical structure	Changes storage
More difficult	Easier

Physical changes occur in storage (e.g. adding an index, changing file organization, moving data, using compression) — the logical structure stays unchanged.

QUESTION 8

DBMS Software Components

DBMS Software Components



Query Processor and Storage Manager



Query Processor

- Reads SQL queries
- Checks query syntax
- Optimizes queries
- Executes queries
- Retrieves results



Storage Manager

- Stores database files
- Retrieves data
- Manages memory
- Maintains indexes
- Controls physical storage

Transaction Management



Transaction Manager

- Controls database transactions
- Maintains consistency
- Handles concurrent users
- Supports recovery after failure



ACID Properties

- Atomicity — complete or cancel
- Consistency — data remains valid
- Isolation — transactions do not interfere
- Durability — completed data remains stored

QUESTION 9

Strong and Weak Entities

Strong Entity

S

Characteristics

- Has its own primary key
- Can exist independently
- Represented using a single rectangle

E

Example

- STUDENT (Student_ID, Name, Department)
- Student_ID uniquely identifies every student

Weak Entity



- Does not have a complete primary key
- Depends on a strong entity
- Represented using a double rectangle
- Uses an identifying relationship
- A dependent cannot exist without an employee

Comparison of Strong and Weak Entities

Feature	Strong Entity	Weak Entity
Primary key	Has its own key	Depends on owner key
Existence	Independent	Dependent
Symbol	Single rectangle	Double rectangle
Example	Student	Dependent

Strong entities exist independently, while weak entities depend on strong entities.

QUESTION 10

Retail Chain EER Design

Retail Chain Management System

P

Purpose — Manage:

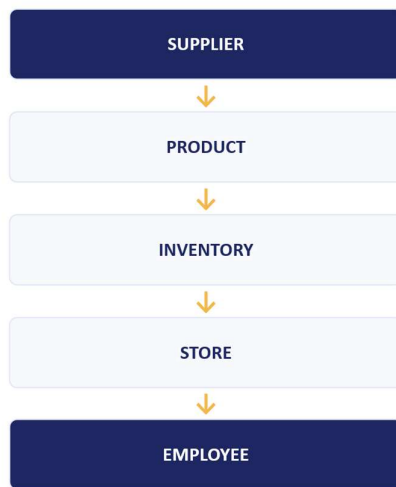
- Multiple stores
- Products
- Employees
- Customers
- Suppliers
- Inventory
- Sales

E

Main Entities

- Store
- Product
- Employee
- Customer
- Supplier
- Sale
- Inventory

Retail Chain EER Design



- Supplier supplies Product
- Product is stored in Inventory
- Inventory is maintained by Store
- Store employs Employee
- Separately: CUSTOMER makes SALE, SALE contains PRODUCT

Retail EER Specialization and Conclusion



Conclusion



Key Takeaways

- DBMS provides secure and efficient data management
- Data models define how information is organized
- ER and EER models support effective database design
- Data independence improves flexibility
- DBMS components manage queries, transactions, and storage
- DBMS is widely used in real-world systems